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Investigation of microwave dielectric properties of biodiesel components

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HIGHLIGHTS

- Dielectric properties of biodiesel precursors were measured.
- The effect of temperature and frequency on the dielectric properties was studied.
- Temperature and frequency has significant effect on the dielectric properties.
- The effect varies with the polarity and conductivity of substance.
- Catalyst has a significant effect on dielectric properties.

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ABSTRACT

Advanced microwave technology has the potential to significantly enhance the biodiesel production process. Knowledge of dielectric properties of materials plays a major role in microwave design for any process. Dielectric properties (ε' and ε'') of biodiesel precursors: soybean oil, alcohols and catalyst and their different mixtures were measured using a vector network analyzer and a slim probe in an open ended coaxial probe method at four different temperatures (30, 45, 60 and 75 °C) and in the frequency range of 280 MHz to 4.5 GHz. Results indicate that the microwave dielectric properties depend significantly on both temperature and frequency. Addition of catalyst significantly affected the dielectric properties. Dielectric properties behaved differently when oil, alcohol and catalyst was mixed at room temperature before heating and when the oil and the alcohol catalyst mixture was heated separately to a pre-determined temperature before mixing. These results can be used in designing microwave based transesterification system.

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1. Introduction

Microwave dielectric heating is a well-established procedure used in industrial processing of food and other materials, such as, ceramics, rubbers, and plastics (Meredith, 1998; Metaxas and Meredith, 1993) as well as for home cooking and heating. Major advantages of microwave processing include rapid heating, safety and, in certain applications, improved energy efficiency. The advantages stem from the direct molecular level interaction leading to volumetric heating. The interaction of electromagnetic waves with matter are characterized by two physical quantities – complex dielectric permittivity, ε and complex magnetic susceptibility, μ (Stuerga, 2008). The latter is relatively unimportant in relation to most biological and agricultural materials that exhibit little magnetic susceptibility. Thus, dielectric heating is induced by the electric field via two mechanisms: dipolar rotation and ionic conduction.

For the first mechanism, a dipole, when exposed to an electric field, will try to align itself with the corresponding field polarity. When the applied field is rapidly changing, the dipolar molecules try to realign to the new direction (also rapidly) and at certain frequencies, the dipole ends up spinning around. In this effort of alignment, the molecules rasp with each other causing friction and subsequent heating. In ionic conductance, charged ions oscillate through the solution under the influence of the same applied electric field. As the electric field direction changes, the ions slow down and change direction. In the process, this motion causes collisions resulting in conversion of kinetic energy into heat energy. The interaction phenomena induced between the electromagnetic waves and a dielectric substance can be divided into two major factors: dielectric constant ε' – the storage of electromagnetic energy in the material, and dielectric loss factor ε'' – the thermal conversion of electromagnetic energy occurring as explained above. The relation between permittivity ε, ε' and ε'' is given by the complex equation (Gabriel et al., 1998):

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$$\varepsilon = \varepsilon' - j\varepsilon'' \quad (1)$$

Dependence of dielectric permittivity on frequency and temperature is a complex phenomenon. Frequency dependence of pure polar substances can be expressed using Debye's equation (Hippel, 1954), or, for non-pure polar materials, using Cole–Cole equation (Nelson, 1973). At low frequencies, the electric field is reversed at a slower pace, allowing more time for molecules to align and to store the applied charge in the dipoles, increasing ε'. As frequency increases, the field oscillation is rapid, not allowing molecules to stabilize in the field and store energy, decreasing ε'. The relaxation spectrum depends on various material parameters such as temperature, viscosity, molecular weight and concentration.

As Debye's and Cole–Cole equations are valid only for pure substances, several mixing rules have been derived based on volumetric fractions to calculate the dielectric properties of heterogeneous and homogenous mixtures including Landau–Lifshitz–Looyenga equation (Dube, 1970; Landau and Lifshits, 1960), Bottcher equation (Bottcher, 1945), and Bruggeman equation (Yang et al., 2010). These mixing rules are most commonly applied to non-reacting and powdered mixtures and cannot be readily applied to liquids (Sihvola, 2002) except Bruggeman equation, which is based on the component composition in the mixture and can be applied to liquid mixtures. In case of reacting mixtures, the volume, density, temperature and concentration of reactants and products, as well as that of intermediate products should be accounted for, since these parameters change as the reaction proceeds. The process of analytically calculating the dielectric properties of reacting mixtures at various temperatures is complex, as too many parameters have to be considered. Some studies attempted to calculate the dielectric properties of liquid mixtures by modifying the above mentioned mixing rules with respect to the component properties, but only for the most simple cases (Harvey and Prausnitz, 1987), and are not applicable to the case presented here.

Another factor that can be used to explain the dielectric heating is the loss tangent (or loss angle δ ration of ε'' to ε'), which depends on relaxation time (τ). The torque exerted by the electric field induces rotation in the polar molecules, but the rate at which molecules orient could be slower compared to applied electric field. This delay between electromagnetic stimulation and molecule response, e.g. τ, is responsible for ε''. An increase in ε'' is observed when the equation $\omega = 1/\tau$ is satisfied, where $\omega = 2\pi/T$, with T being the oscillation period (Stuerga, 2008). If a solvent's τ is higher than about 65 ps at 2450 MHz, there is an increase in loss tangent with temperature, resulting in higher rate of conversion of electromagnetic energy into thermal energy (Baghurst and Mingos, 1992). This phenomenon is responsible for the localized superheating of organic solvents in the microwave region (Baghurst and Mingos, 1992). If τ is many times greater than the period T, polarization is barely able to develop and ε'' is very small. At low frequencies, where τ is considerably smaller than T, the polarization follows (lags behind) the field and ε'' is also small because the number of reorientations per unit time is small. Loss tangent for methanol and ethanol are 0.659 and 0.941, respectively at 915 MHz, and the relaxation times for methanol and ethanol are 51.5 and 170 ps, respectively (Kappe, 2008; Pace et al., 1968). Relaxation time is temperature dependent and decreases as the temperature is increased; it is also viscosity dependent (Metaxas and Meredith, 1993);

$$\tau = \frac{4\pi r^3 \mu}{k_b Temp}. \quad (2)$$

where; r is gyration radius of the molecule (m), μ is the viscosity of solution (Pa·s), *Temp* is absolute temperature (K) and *k_b* is Boltzmann's constant ($1.3806504 \times 10^{-23}$ J/K). Longer τ's indicate low heat absorption due to slower response of molecules to the changing electric field.

Although the advantages of microwave heating are long known, it was employed in organic chemistry only starting in mid 80's (Palacios et al., 1996). The application has now gained momentum in the chemical industry because of its rapid and targeted heating capabilities. Microwaves are introduced to the reaction site remotely and can pass through the vessel wall without heating the vessel. As such, microwaves interact volumetrically only with the reactants and solvents, and, for a well-designed apparatus, heating can be uniform without causing the decomposition of reactants or products. One promising application of this technology in the bioenergy area is production of biodiesel, a promising alternative for petroleum fuel, via transesterification of vegetable oil in the presence of catalyst at mild temperatures. The use of alcohols as a solvent and sodium hydroxide as a catalyst gives an unique combination, with the presence of both polar and ionic components in the mixture speeding up the heating process. Variables affecting the performance of this process are microwave frequency, design of cavity, power level, initial temperature, and dielectric properties of reacting materials (Terigar et al., 2010).

Whereas most parameters are independent of the reacting material and can be selected for each specific application, dielectric properties are material specific. Therefore knowledge of dielectric properties of material is beneficial for process design calculations, execution and control of microwave heating. Dielectric properties help in predicting electromagnetic field distribution, heating rates, temperature profiles for the reactions and overall understanding of the mechanism of microwave dielectric heating. Distribution of electric field in the system is heavily influenced by the dielectric properties and cavity geometry. Electric field intensity dictates the heating rates and magnitude of molecular rotation and mixing. This design is of particular importance while designing and building resonant cavities with specific dimensions required for optimum process performance. Dielectric properties are critical in evaluating and optimizing the system performance using numerical modeling approach for this particular multiphysics problem as demonstrated elsewhere (Muley, 2011; Muley and Boldor, 2012). Dielectric properties play an important role in calculating energy efficiency of the reaction process according to (Nelson, 1992);

$$P_{abs} = \sigma E^2 = 2\pi f \varepsilon_0 \varepsilon'' E^2. \quad (3)$$

where, P_{abs} is power absorbed (W/m^3), σ is the conductivity (Siemens/m), f is frequency (Hz), ε_0 is the dielectric constant of the vacuum, and E is electric field intensity (V/m).

The goal of this study was to determine and compare, for the first time to our knowledge, the dielectric properties of biodiesel precursors (oil, alcohol and catalyst) and their mixtures at four different temperatures and in a frequency range of 280 MHz to 4.5 GHz. This approach provides a new, comprehensive set of data, not available until now, related to the dielectric behavior of biodiesel precursors and its constituents and mixtures of very complex mixture of reactants (polar, non-polar and ions), reaction products and intermediates at different temperatures and microwave frequency ranges.

2. Methods

Soybean oil was purchased from a local grocery store whereas ethanol (200 proof, ACS/US grade) and methanol (200 proof, ACS/US grade) were purchased from Pharmco-AAPER (Brookfield, CT). Anhydrous sodium hydroxide flakes were supplied by Fisher Scientific (Pittsburgh, PA).

Dielectric properties of the mixtures were determined using an Agilent ENA series E5071C Network Analyzer and Agilent 85070E dielectric probe kit (Agilent Technologies, Inc. Santa Carla, CA) using a slim form open-ended probe method in a 201-point

frequency sweep from 280 to 4500 MHz. Frequency range was selected based on the limits of measurement system (Agilent 85070E dielectric probe kit is suitable for frequencies above 200 MHz). The network analyzer was controlled by Agilent 85070E dielectric kit software (Agilent Technologies, Inc. Santa Carla, CA) and calibrated using the 3-point method (short-circuit, air and water at 25 °C).

As biodiesel can be produced using either alcohols (methanol or ethanol) it was important to study the dielectric nature of both as well as their different mixtures with oil and catalyst. The dielectric properties of pure components: soybean oil, ethanol and methanol, and their various mixtures (including those with basic catalyst) were measured at four different temperatures of 30, 45, 60 and 75 °C. These temperatures, selected to cover the range from room temperature to alcohol boiling points, was maintained using a constant temperature water bath. Since methanol has a boiling point of 68 °C, its dielectric properties were not measured at 75 °C. Second step involved measurement of dielectric properties of solvent with basic catalyst. Anhydrous sodium hydroxide (0.18% by the weight of oil) was fully dissolved in both ethanol and methanol (Kanitkar et al., 2011) using a magnetic stirrer prior to measurements. The final step involved measurements of dielectric properties of mixture of oil with solvent and the catalyst, as this is the mixture from which biodiesel is prepared. The oil to ethanol ratio was maintained at 1:9, while the oil to methanol ratio was 1:5 (Kanitkar et al., 2011). These measurements were performed for two different scenarios. In each scenario the catalyst was fully premixed with the alcohols.

In the first scenario, oil and the alcohol- catalyst mixture were heated separately and were mixed only when the desired temperature was attained. The dielectric properties were measured immediately after adding oil and ensuring sample homogeneity via stirring. As the data was collected immediately, it gives insight into the dielectric properties of oil solvent mixture before the transesterification reaction was started.

The second scenario involved mixing of oil with the solvent-catalyst mixture at room temperature, followed by heating in a controlled temperature water bath at different temperature with continuous stirring. Measurements were collected as the temperature reached each of the value prescribed above. This methodology provided the dielectric properties at different temperature for a sample undergoing transesterification. The two type of heating methods (while undergoing transesterification and prior to it) will help in understanding the microwave dielectric heating for the biodiesel precursor components and will be useful in designing and optimizing a microwave-based transesterification process.

The data, collected in triplicate, was imported into Microsoft Excel® 2010 (Microsoft Corp., Redmond, WA). Data for each sample was averaged across each replicate measurement and replicate sample, and standard deviations of each sample were calculated. The results were imported into Sigma Plot® 10 (Systat Software Inc., San Jose, CA) for plotting graphs.

3. Results and discussion

3.1. Soybean oil

3.1.1. Dielectric constant

Dielectric constant measured for soybean oil is essentially unchanged throughout the frequency range; however a very slight decrease is observed as the frequency increases (Fig. 1). The dielectric constant values are relatively low (the highest being 3.2), the main reason being the non-polar nature of oil. For comparison, the ε' for a polar solvent such as water is 80.10 (Deshpande et al., 2010). Similar results are found in literature where the dielectric properties of soybean oil were measured at varying frequency and temperatures. For example, the value at 915 MHz and 30 °C of 2.91 obtained in this study compares well to the 2.64–3.00 range presented in literature (Elshami et al., 1992; Pace et al., 1968; Sorichetti, 2011).

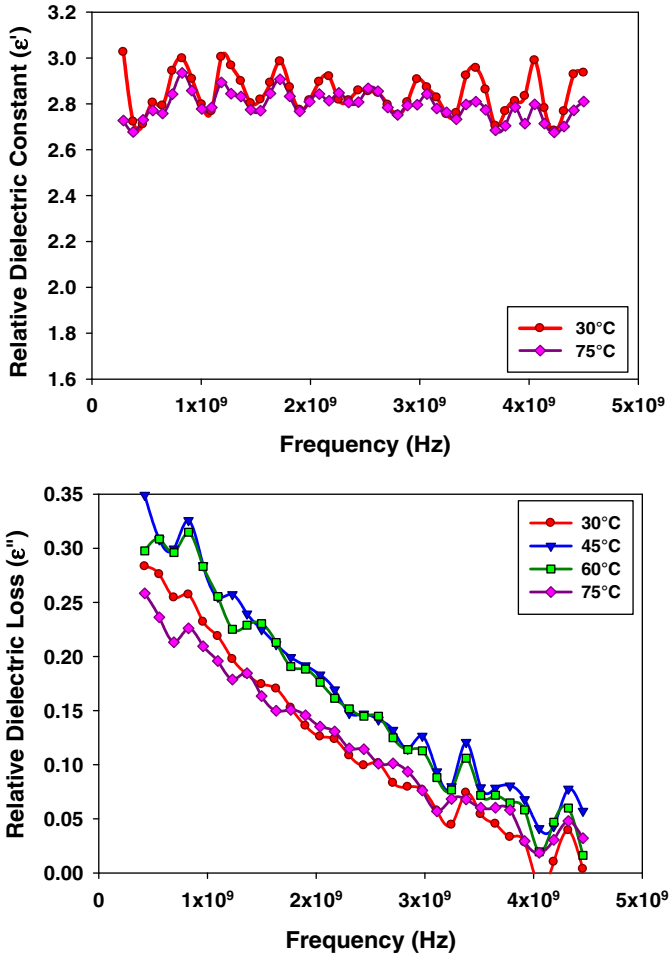


Fig. 1. Dielectric constant of soybean oil at 30 and 75 °C (For clarity of data, only the values obtained at lowest and highest temperature is shown in the graph. Other temperature values essentially overlapped) and dielectric loss of Soybean oil at different temperatures and frequency.

For a frequency range much lower than investigated here, (10^{-5}–1 MHz) the decrease in the ε' values as the frequency increases can be substantial (Hu et al., 2008). As resistance from the viscosity of the medium impedes the process of reorientation, with the rise in frequency the alignment becomes even harder due to inertial effects, causing the decrease in ε' (Pace et al., 1968). Fatty acid composition of oil is another determining factor of ε'. Soybean oil belongs to the linoleic acid class of fatty acids. Dielectric constant for this class of fatty acid noted by Hu et al., 2008., at low frequency (10^{-4}–0.5 MHz) was 3.104–3.127 and decreased as the frequency was increased to 1 MHz. Dielectric constant increases with the increase in degree of unsaturation as the electron rich double bond has a negative charge and tends to oscillates to certain extent, with the changing electric field (Hu et al., 2008). However as it can be observed in our study, the decrease with frequency is relatively small at the higher frequency investigated here (280–4500 MHz).

Dielectric constant of oils decreases with the increasing temperature (Elshami et al., 1992; Sorichetti, 2011). This observation was prominent in literature compared to our results (virtually constant with respect to temperature). A possible explanation is that literature data covered a frequency range at much lower frequencies

(from 10^{-5} to 10^{3} MHz) than those investigated here. As the temperature rises, density decreases; molecular motion increases as well as molecule-to-molecule interaction (Elshami et al., 1992; Pace et al., 1968). These phenomena oppose the polarization process as associated with the orientation of molecules with respect to the changing applied electric field (Sorichetti, 2011), therefore reducing ε'.

3.1.2. Dielectric loss

Dielectric loss of oil for the overall frequency range and at all the temperatures was negligible (max. 0.366 at 468 MHz) at least when compared to polar solvent such as water (6.0 at 460 MHz). The results indicated a clear decrease with the increase in frequency and slight (Fig. 1). For a constant temperature, with the changing frequency, there is sharp decrease in ε''; the high viscosity and inertial effects prevent the molecules (which are non-polar, non ionic and therefore weakly interacting with the field) from reorientation at these frequency values. In literature, data for low frequency (10–1000 MHz) shows an increase in ε'' and loss tangent with increasing frequencies (Sorichetti, 2011).

At a constant frequency, with the increase in temperature, the ε'' varied slightly. As the temperature was increased from 30 to 45 °C and further to 60 °C the ε'' increased; however as the temperature was further increased to 75 °C, the ε'' decreased. This particular behavior of ε'' with respect to temperature was observed by other researchers and it was reported in literature (Pace et al., 1968). At lower temperatures (30 °C in this case), the viscosity is high and so the relaxation time is long, resulting in increased values of ε''.

3.2. Alcohol

3.2.1. Dielectric constant

Ethanol is a polar solvent; it has a permanent dipole moment. The dielectric constant of polar solvents is frequency dependent (Pace et al., 1968) and shows a distinct decrease as the frequency increases. The data shows a decrease in the ε' with the increase in frequency. Similar results were found in the literature where the trend of ε' is similar to the one indicated by our experiments (Pace et al., 1968). The decrease in ε' with frequency is confirmed by Debye's equation (Hippel, 1954). In our case, we studied the changing temperature parameter. It is observed that the rate of decrease of ε' is smaller at higher temperature (Fig. 2(a)). This could be because at high temperatures the kinetic energy of molecules is high; hence the response to the changing electric field is faster and the viscosity is smaller (molecules are further apart), which enhances the realignment process, increasing the dipolar rotation of molecules and ultimately the ε' of the material compared to lower temperature. As temperature increases, a shift in frequency at which ε' starts dropping was observed. A peculiar characteristic

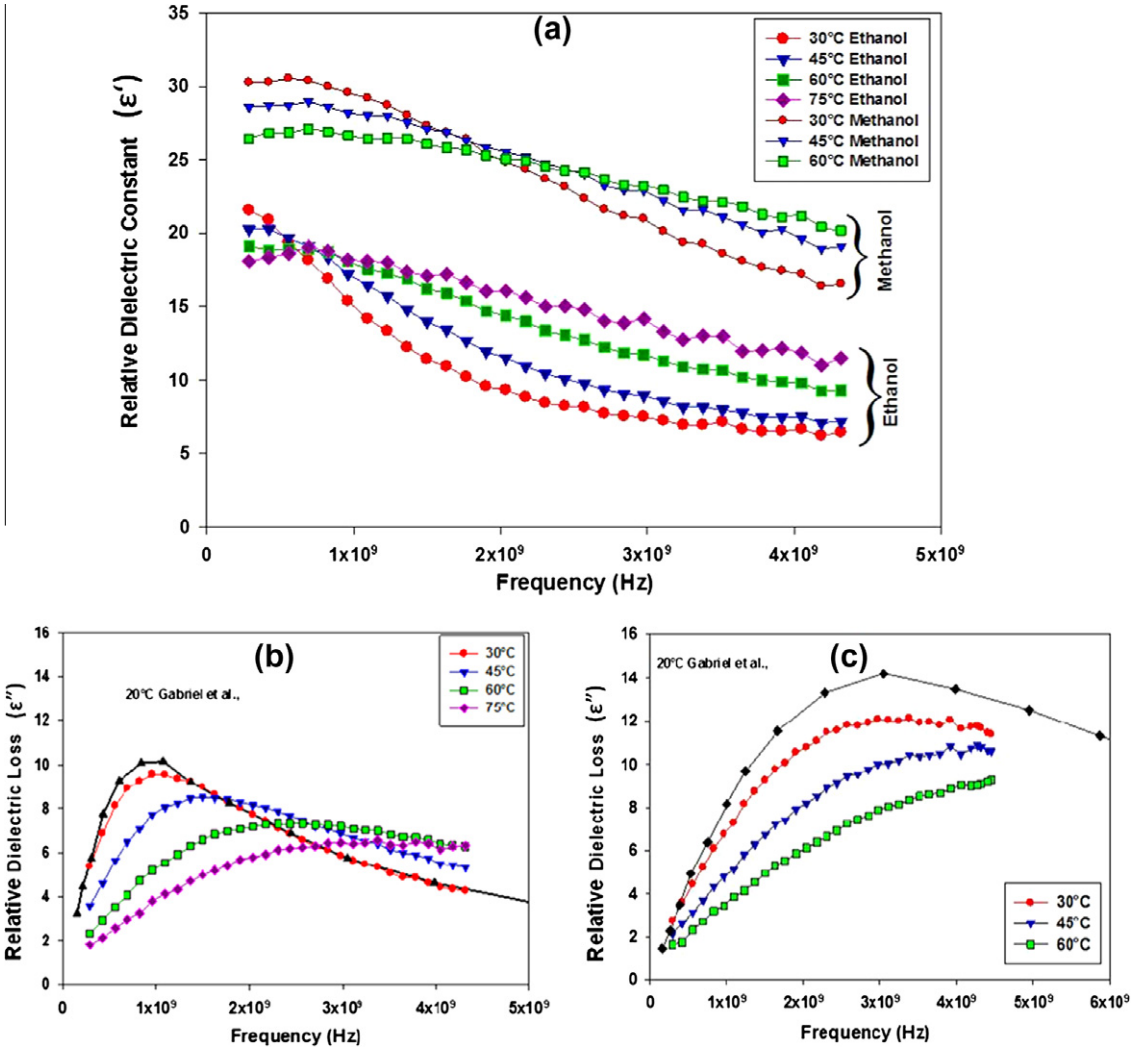


Fig. 2. Dielectric properties of alcohol. (a) Dielectric constant of methanol and ethanol (b) Dielectric loss of ethanol (c) Dielectric loss of methanol.

of a polar molecule; as the temperature rises, dipolar rotation is more prominent causing a shift in the location of where ε' decreases (Terigar et al., 2010).

Similar trends were observed for methanol. The graph shows a decreasing trend for ε'. However, methanol had higher ε' values for a given frequency compared to that of ethanol, ranging between the highest values of 31.0 at a frequency of 512 MHz to almost half (18.0) for the highest frequency of 4500 MHz (Fig. 2(a)). The difference between the highest and lowest value of ε' (31–18) is again highest for the lowest temperature of 30 °C. By comparison, at 60 °C the decrease is from 28 to 23 for the same frequency range. The rate of decrease of ε' with frequency was smaller for methanol compared to ethanol because, as the number of CH_2 groups in R–OH chain increases, the ε' decreases. When a hydrogen atom bonds with an oxygen atom, the electron distribution in the newly formed molecule alters and induces a dipole moment that changes the polarization. But a combination of carbon and hydrogen in the hydrocarbon portion of a molecule is always non-polar. On the other hand, methanol is more polar because of the shorter distance between the poles and its smaller molecular size, which allows it to rotate and realign faster (smaller molecules also have a higher dipole moment).

3.2.2. Dielectric loss

Dielectric loss for both the alcohols was frequency and temperature dependent. Loss factor increased with increasing frequency until it reached a peak, and then a gradual decrease was observed. With the increasing temperature, the ε'' peak shifted towards the higher frequency value. For ethanol, the ε'' peak at 30 °C is observed around 1000 MHz (Fig. 2(b)), whereas for methanol at 30 °C the peak falls between 3000 and 4000 MHz (Fig. 2(c)). At the higher temperatures, the peak seems to fall outside the range measured in this study for methanol. This trend is consistent with the data reported in literature, in which the ε'' peak for methanol measured at 20 °C is observed at 3000 MHz (Gabriel et al., 1998). The highest value of ε'' for both alcohols was observed at lowest operating temperature of 30 °C. At lower frequencies (<1000 MHz), ε'' values for methanol were lower than those of ethanol. As the frequency increased, the decreasing values for ethanol became smaller than those of methanol, which, for the most part, did not reach their peaks in the frequency range investigated here.

Relaxation frequency is defined as the frequency at which the ε'' for a given substance is highest. In order to calculate a more precise value of the relaxation frequency, we plotted the first derivative of ε'' factor of alcohol with respect to the frequency $(d\varepsilon''/df)$ (Fig. 3(a) and (b)). At lower temperature (30 °C) the relaxation frequency of ethanol is 1005.6 MHz and that for methanol is 3463 MHz. As the temperature increases, the relaxation frequency rapidly increases [Table 1]. Hence, at a lower working frequency (below 2000 MHz), the dielectric heating will be higher for ethanol at lower temperatures. At higher working frequency (above 2000 MHz), the thermal conversion of electromagnetic energy is predominant at higher temperatures. Also, at frequencies below 1500 MHz, the ε'' is higher for ethanol that for methanol. Thus, ethanol would cause better heating at 915 MHz compared to methanol.

Similar results have been reported in the literature confirming the dependence of ε'' on temperature (Liao et al., 2001) and on frequency (Pace et al., 1968). As Federal Communications Commission (FCC) allows only certain microwave frequencies of operation (915 and 2450 MHz) for industrial, medical and scientific applications, it is impractical to build a system with variable frequency in order to match the temperature behavior of material undergoing processing. Therefore, in order to design a process, one needs to choose an optimum frequency of operation from among those permitted in regulations. The data collected during this study provides the exact behavior of alcohols at different temperatures in order to optimize the heating process and select the most suitable frequency at which high heating rates are maintained throughout

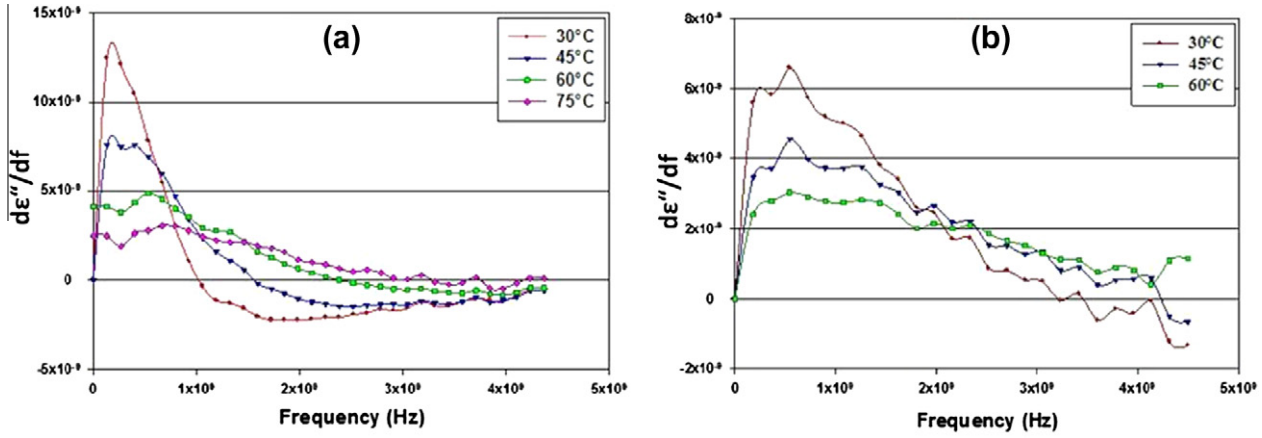


Fig. 3. First order derivative of dielectric loss with respect to frequency (a) Ethanol, (b) Methanol.

Table 1
Relaxation time and viscosity of ethanol and methanol at different temperatures.

Temperature (K)	Ethanol ($r = 4.8 \times 10^{-10}$ m)				Methanol ($r = 2.65 \times 10^{-10}$ m)			
	Relaxation frequency (MHz)	Relaxation time (s)	Viscosity (cP)	Viscosity literature (cP)	Relaxation frequency (MHz)	Relaxation time (s)	Viscosity (cP)	Viscosity literature (cP)
303.15	1005.6	1.58×10^{-10}	0.2	0.1	3463	4.59×10^{-10}	0.013	0.05
318.15	1540	1.03×10^{-10}	0.1	0.09	4231	3.76×10^{-10}	0.012	0.04
333.15	2378	6.69×10^{-11}	0.13	0.06	5830	2.73×10^{-10}	0.01	0.03
348.15	3749	4.24×10^{-11}	0.08	0.05	-NA-	-NA-	-NA-	-NA-

the temperature range. For a practical purpose, this selection is between 915 and 2450 MHz as equipment is usually build only for these frequencies.

Relaxation frequency depends on viscosity of the material. In order to confirm our results, we calculated the viscosity of ethanol and methanol [Eq. (2)] based on the relaxation frequency that was obtained from the first order derivative of ε''. Viscosities obtained at different temperatures are given in Table 1. The results are in close agreement with the standard viscosity data for ethanol but varies for methanol (Lide, 2000).

3.3. Alcohol-catalyst

The dielectric behavior of any electrolytic solution (such as alcohol and basic catalyst) is complicated. Ions have a strong electric field; hence ions affect the molecular interaction of solvents. Although the amount of NaOH added was very low (0.18% by the weight of oil for transesterification, corresponding to 0.066 g catalyst in 19.3 ml ethanol and 0.091 g catalyst in 10.235 ml methanol), its effect on the dielectric properties was quite spectacular.

3.3.1. Dielectric constant

The dielectric constant for alcohol-catalyst mixture decreased with increasing frequency, following a similar trend with what is observed for pure alcohols (Fig. 4(a)). For a fixed frequency, the ε' increased with a rise in temperature. Overall ε' is smaller for the alcohol-catalyst mixture than for pure alcohol by approximately 20%.

Depending on factors such as ion concentration, solvent properties and the association and dissociation of ions, ε' may stay constant or decrease. In this case the decrease in ε' compared to pure alcohol is primarily due to ion solvation (Hasted and Roderick, 1958). Solvation of ions takes place when the polar solvent molecules orient towards the ion in response to ion's electrostatic field. This association with ions causes the system to stabilize by forming a molecular solvation shell around each ion. The solvent molecules that are bound in this shell cannot orient as easily by the applied electric field; therefore they do not generate effective dipole moment, decreasing polarization and hence reducing the ε'. Interactions between ions and solvent dipoles lead to dielectric saturation of solvent molecules in the ion's electric field by inhibiting the free rotation of the solvent molecules (Hasted and Roderick, 1958).

3.3.2. Dielectric loss

The influence of added ions was even more pronounced on ε'', which not only increased significantly, but also changed the frequency dependence (Fig. 4(b) and (c)) toward an asymptotic decrease as frequency increased. A significant difference was observed with the changing temperature for both mixtures. Unlike pure alcohol (for which a peak was observed), ε'' for alcohol-catalyst mixture decreased monotonously with frequency. For a partic-

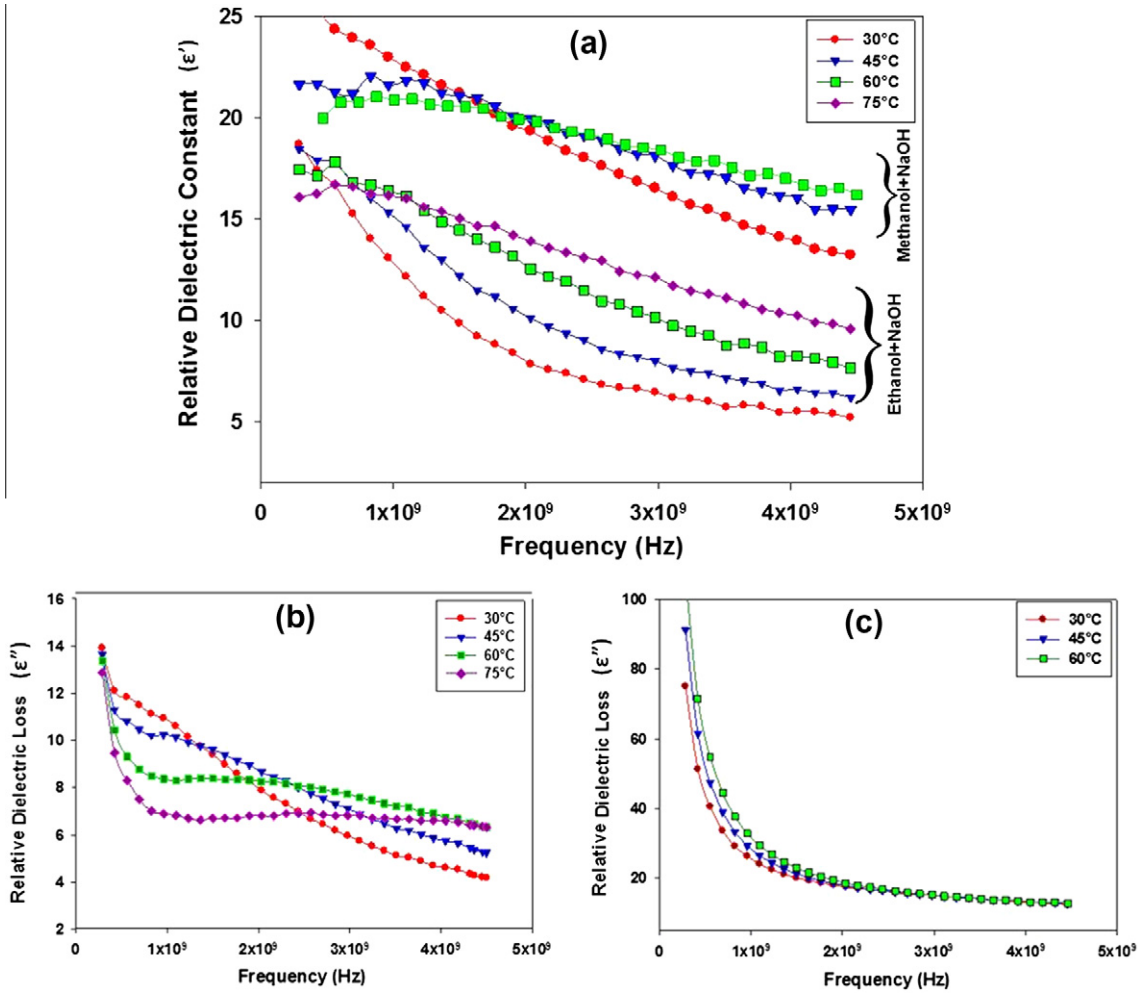


Fig. 4. Dielectric properties of alcohol- catalyst mixture. (a) Dielectric constant of alcohol + NaOH (b) Dielectric loss of ethanol + NaOH (c) Dielectric loss of methanol + NaOH. Note different scale on *y*-axis for (b) and (c).

ular frequency, the individual ε'' value was much higher for the mixture when compared with pure alcohols. Addition of small amount of electrolyte increases the conductivity (especially at lower frequencies) and ultimately increases the dielectric heating. Generally, solutions with higher alkali concentration are more effective in absorbing microwave energy and converting the stored energy into heat. Since ionic conduction is a major factor for thermal effects in microwave processes, a system with a higher concentration of NaOH should exhibit more ionic conduction in response to microwave energy. In our study, in spite of the very low NaOH concentration, there was a significant increase in ε''. While some studies have been performed on dielectric properties of alcoholic electrolytic solution, we did not find any literature data describing the dielectric properties of alcohol- NaOH solution (Hasted and Roderick, 1958). When compared to literature data on NaOH dielectrics in different solvents (generally water), the trends for ε' were similar to those presented here. But for ε'', literature data reported a rise in values with increase in frequency and increased concentration of NaOH (Kelly and Salomon, 1969; Kessler and Mariani, 1967).

Determining the temperature dependence behavior of an electrolyte solution is a very complicated process. For ethanol-catalyst mixture, a decrease in ε'' with temperature was observed at lower frequencies (up to 2450 MHz). As the frequency increased above 2450 MHz, the ε'' increased with temperature. The ε' and ε'' values at 915 and 2450 MHz, which are the frequencies allocated by the FCC for industrial, scientific and medical application are tabulated (Table 2). At 915 MHz, the loss tangent for methanol-catalyst mixture is higher than that for ethanol-catalyst mixture. Higher loss tangent values ensure higher heating rates (Pace et al., 1968).

3.4. Oil, alcohol and catalyst mixture

3.4.1. Oil ethanol catalyst (scenario one)

3.4.1.1. Dielectric constant. The dielectric constant was observed to decrease with increasing frequency and temperature for both scenarios. However, the value of ε' at an individual frequency for scenario one was higher compared to scenario two for all temperatures (Fig. 5(a) and (b)). These higher values were due to mixing the components already at the ascribed temperatures, with little time for them to undergo the transesterification reaction. A higher amount of ethanol was available, which increased solution polarity. At the lowest temperature of 30 °C, the chances of reaction were extremely small, thus higher ε' values are observed compared to other temperatures (Table 2).

3.4.1.2. Dielectric loss. Decrease in dielectric loss was observed with the rise in temperature at lower frequencies (Fig. 5(c)). Again, with the increase in frequency, the polar molecules lag behind in response to the applied electric field. Temperature dependence was not as strong as frequency increased; mainly due to reduction in mobility of molecules. As the reaction proceeds, polar components (in this study alcohol) are utilized and form intermediate (mono and diglycerides) and final products. Many of the intermediate products formed during transesterification reaction are non-polar compounds (or marginally polar). These can be termed as inactive components for microwave heating, with unreacted ethanol acting as an active component. Thus dielectric loss value at individual frequencies was higher for scenario one than scenario two because of the presence of active polar molecules which were the result of incomplete reaction.

3.4.2. Oil, ethanol and catalyst (scenario two)

3.4.2.1. Dielectric constant. As mentioned earlier, ethanol-catalyst solution was well mixed with oil in the ratio of 1:9. The ε' was observed to decrease with the increasing frequency and temperature especially at 30 °C. At other temperatures, the decrease in ε' was less pronounced compared to 30 °C (Fig. 5(b)). As a function of temperature, this is in contrast with the results obtained for pure solvents and the ethanol-catalyst solution. For the pure solvents the ε' increased with the increase in temperature (Table 1). The decrease in ε' with temperature for this mixture was mainly due to low values of ε' for oil and, secondary to the fact that as the temperature increases, oil is converted into biodiesel. At lower temperature the transesterification reaction is not started, thus polarity of ethanol remains dominant. As the temperature is increased, the reaction proceeds and the concentration of polar components are significantly reduced. The product formed, which is essentially either biodiesel (esters of free fatty acids) or the reaction intermediates (monoglycerides and diglycerides) are less polar in nature. As the polarity of the solution is decreased, the ε' decreases. The dielectric properties of oil alcohol catalyst mixture at 915 and 2450 MHz are given in (Table 3).

3.4.2.2. Dielectric loss. At the lower temperatures of 30 and 45 °C, a peak of the ε'' was observed at lower frequencies; a peak was not

Table 2
Values for dielectric constant and dielectric loss of pure components and alcohol-catalyst mixture at 915 and 2450 MHz.

Component	Temperature	915 MHz			2450 MHz		
		ε'	ε''	*Tan* δ	ε'	ε''	*Tan* δ
Soybean oil	30 °C	2.90 ± 0.01	0.23 ± 0.0001	0.079	2.86 ± 0.03	0.10 ± 0.003	0.035
	45 °C	2.97 ± 0.01	0.28 ± 0.006	0.096	2.85 ± 0.02	0.14 ± 0.003	0.052
	60 °C	2.99 ± 0.03	0.28 ± 0.007	0.094	2.95 ± 0.04	0.14 ± 0.007	0.049
	75 °C	2.85 ± 0.21	0.21 ± 0.01	0.075	2.81 ± 0.05	0.11 ± 0.009	0.040
Ethanol	30 °C	15.85 ± 0.18	9.54 ± 0.11	0.602	8.21 ± 0.11	6.83 ± 0.08	0.831
	45 °C	17.68 ± 0.14	7.57 ± 0.25	0.428	10.02 ± 0.02	7.63 ± 0.03	0.762
	60 °C	18.22 ± 0.04	5.06 ± 0.06	0.278	13.01 ± 0.05	7.32 ± 0.02	0.562
	75 °C	18.35 ± 0.05	3.63 ± 0.07	0.198	14.98 ± 0.04	6.20 ± 0.08	0.413
Methanol	30 °C	29.51 ± 0.14	6.55 ± 0.16	0.222	23.03 ± 0.04	11.56 ± 0.13	0.502
	45 °C	28.34 ± 0.26	4.63 ± 0.22	0.163	24.33 ± 0.03	9.13 ± 0.02	0.375
	60 °C	26.84 ± 0.39	3.30 ± 0.10	0.123	24.25 ± 0.04	6.95 ± 0.02	0.286
	75 °C	-NA-	-NA-	-NA-	-NA-	-NA-	-NA-
Ethanol + NaOH	30 °C	13.57 ± 0.28	11.04 ± 0.09	0.814	7.03 ± 0.08	6.95 ± 0.11	0.988
	45 °C	15.76 ± 0.27	10.25 ± 0.25	0.650	8.97 ± 0.03	7.96 ± 0.27	0.888
	60 °C	16.50 ± 0.01	8.41 ± 0.06	0.509	11.39 ± 0.03	8.06 ± 0.02	0.707
	75 °C	16.32 ± 0.08	6.92 ± 0.048	0.424	13.04 ± 0.06	6.89 ± 005	0.528
Methanol + NaOH	30 °C	23.34 ± 0.15	27.12 ± 0.32	1.162	17.99 ± 0.05	16.24 ± 0.002	0.903
	45 °C	21.84 ± 0.03	30.42 ± 0.48	1.392	19.10 ± 0.12	16.32 ± 0.07	0.854
	60 °C	20.80 ± 0.03	34.18 ± 0.21	1.643	19.22 ± 0.01	16.74 ± 0.02	0.871
	75 °C	-NA-	-NA-	-NA-	-NA-	-NA-	-NA-

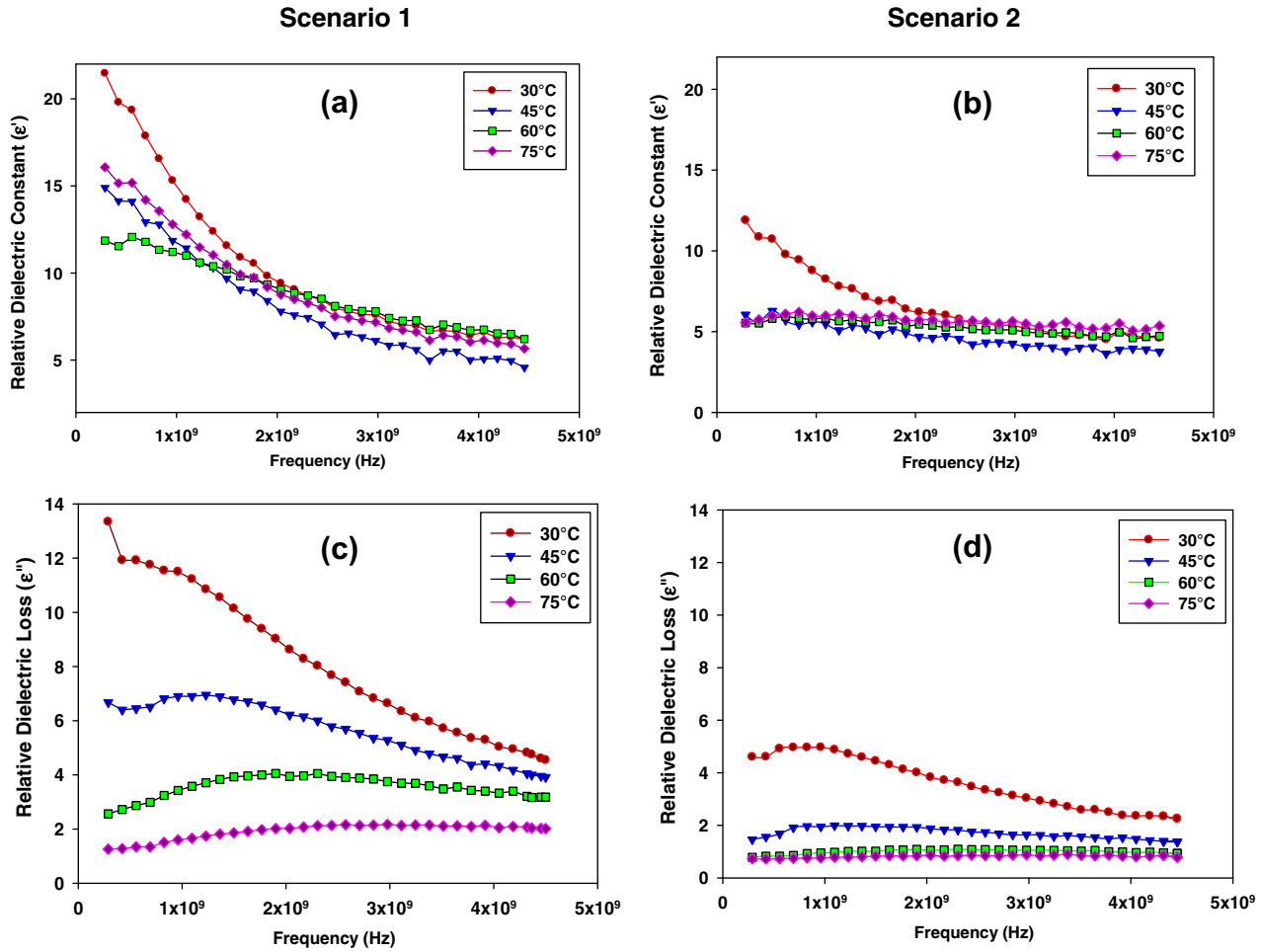


Fig. 5. Dielectric properties of oil, ethanol and catalyst mixture (a) Dielectric constant ε' for scenario one; (b) Dielectric constant ε' for scenario two; (c) Dielectric losses ε'' for scenario one; (d) Dielectric loss ε'' for scenario two.

Table 3
Values of dielectric constant and dielectric loss at 915 and 2450 MHz for oil alcohol catalyst mixture (Scenario 1 & 2).

Component	Temperature (°C)	915 MHz		2450 MHz	
		ε'	ε''	ε'	ε''
Oil + EtOH – NaOH Scenario 1	30	15.78 ± 0.13	11.55 ± 0.09	8.38 ± 0.07	7.65 ± 0.09
	45	12.23 ± 1.64	6.90 ± 1.56	6.98 ± 0.07	5.70 ± 0.1
	60	11.19 ± 0.01	3.38 ± 0.03	8.47 ± 0.04	3.93 ± 0.03
	75	8.13 ± 0.33	1.58 ± 0.18	7.94 ± 0.18	2.77 ± 0.02
Oil + EtOH – NaOH Scenario 2	30	9.13 ± 0.17	5.00 ± 0.23	5.72 ± 0.08	3.46 ± 0.01
	45	5.61 ± 0.09	1.97 ± 0.05	4.47 ± 0.06	1.75 ± 0.02
	60	5.88 ± 0.05	0.94 ± 0.01	5.25 ± 0.03	1.08 ± 0.01
	75	6.20 ± 0.007	0.72 ± 0.02	5.61 ± 0.02	0.85 ± 0.01
Oil + MeOH + NaOH Scenario 1	30	18.61 ± 0.08	21.03 ± 0.18	21.03 ± 0.05	12.38 ± 0.03
	45	19.84 ± 2.3	26.40 ± 4.32	26.40 ± 0.24	14.28 ± 0.24
	60	10.55 ± 0.56	19.43 ± 2.0	19.43 ± 0.75	9.43 ± 0.99
Oil + MeOH + NaOH Scenario 2	30	11.11 ± 0.16	12.05 ± 0.39	8.81 ± 0.17	6.75 ± 0.1
	45	15.64 ± 0.19	22.43 ± 0.91	13.34 ± 0.33	11.69 ± 0.41
	60	14.38 ± 0.09	22.75 ± 0.02	12.65 ± 0.9	10.97 ± 0.1

observed as the temperature was further increased (Fig. 5(d)). As the frequency increased, ε'' decreased. At lower temperatures the solution is still polar and the ions are free to move about in the solution, hence the higher ε''. Once the reaction proceeds, Na^{++} ions participate in the reaction and thus are not as free to move about. The peak at 30 °C is similar to the peak of ethanol at 30 °C observed at a lower frequency of 1000 MHz but with a lower value of 5.0 compared to 9.5 for pure ethanol, mainly due to the presence of non-polar components (oil) in the mixture. The stoichiometric ratio for the reaction is 1:3 oil to alcohol, however as the reaction is reversible, excess alcohol is added to ensure the completion of reaction. The molar ratio we used for oil to ethanol was 1:9. Stoichiometrically, 6.0 mol of ethanol remain unreacted. This excess ethanol gives the solution a polar nature and hence the ε'' is increased with the increase in temperature. For the higher temperatures of 60 and 75 °C, conductivity (which is another

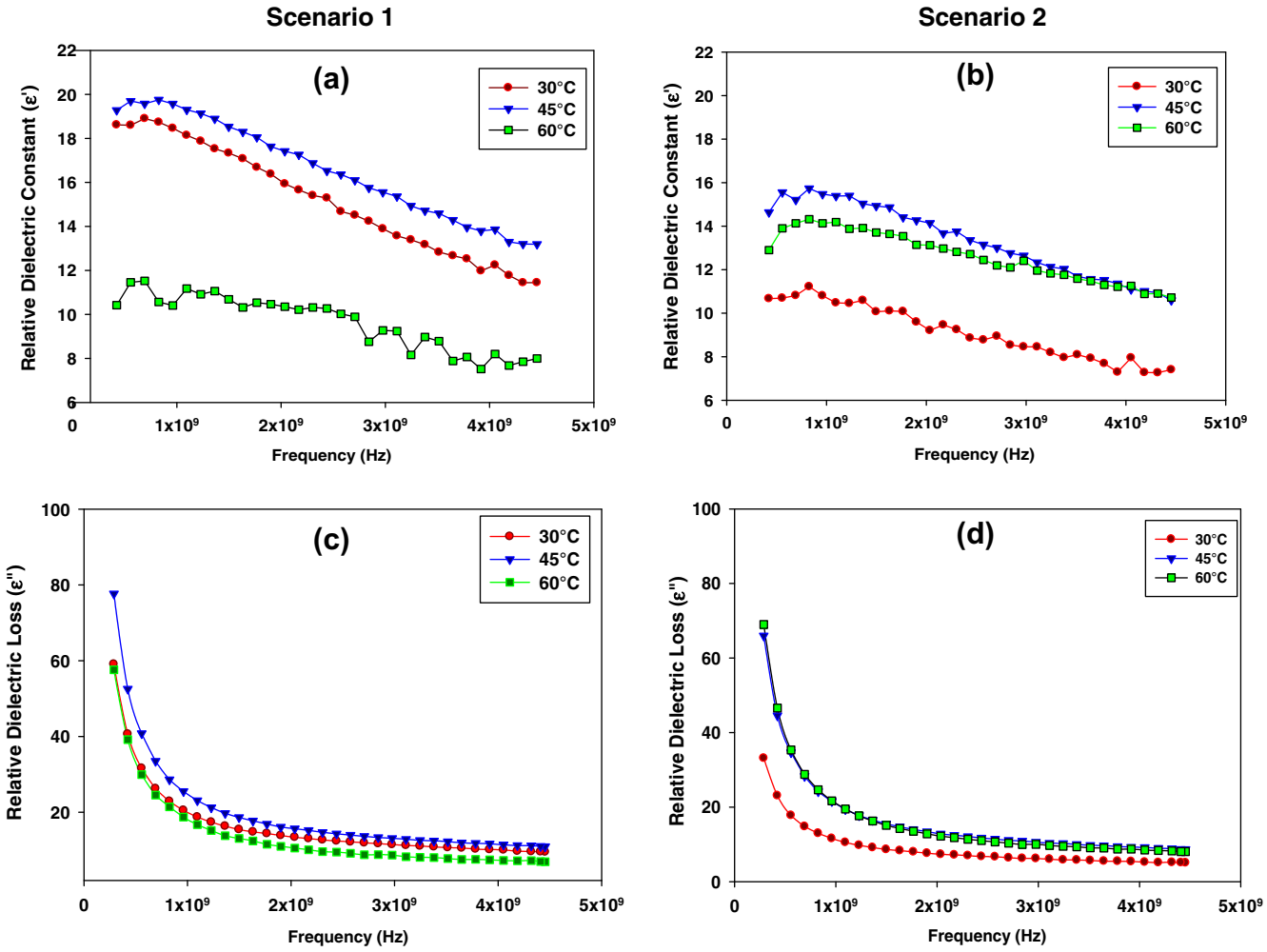


Fig. 6. Dielectric properties of oil methanol- catalyst mixture (a) Dielectric constant for scenario one, (b) Dielectric constant for scenario two, (c) Dielectric losses for scenario one, (d) Dielectric loss for scenario two.

representation of ε'', see Eq. (3)) remained constant throughout the frequency range as the reaction was close to completion. The decrease in ε'' could be because of the less effective dipole moment of polar alcohol molecules due to their association with the non-polar molecules (oil, reaction intermediates) (Kappe, 2008).

3.4.3. Oil methanol catalyst (scenario one)

3.4.3.1. Dielectric constant. Dielectric constant for oil and methanol + catalyst mixture was relatively constant at lower frequency but decreased gradually as the frequency was increased (Fig. 6(a)). Dielectric constant also changed with temperature as for ethanol-oil mixture. This difference was apparent at 60 °C as with the increasing temperature, the reaction proceeded towards completion.

3.4.3.2. Dielectric loss. For oil-methanol mixture in scenario 1, a spectacular decrease in the ε'' values from a very large value was observed at lower frequencies (until the frequency reached 1500–1600 MHz) after which the decrease is much less pronounced (Fig. 6(b)). This behavior is observed for any given temperature. Similar trends were observed for methanol-catalyst solution but the overall values were lower with the addition of oil. Separate heating of oil and methanol-catalyst solution reduces the probability of the mixtures undergoing transesterification reaction. Nevertheless, at high temperature there could be a possibility of reaction taking place hence, a decrease in ε'' was noticed as the temperature was increased.

3.4.4. Oil, methanol catalyst (scenario two)

3.4.4.1. Dielectric constant. The dielectric constant changed with temperature for oil–alcohol + catalyst mixture (Fig. 6(c)). This was true for ethanol–oil at all the temperatures and it also stands true for methanol–oil mixture except at 30 °C. This exception could be because at 30 °C, ion solvation takes place, reducing the ε' of the mixture. As the temperature rises to 45 °C, the catalyst ions participate in the reaction reducing the effect of ion solvation; hence the rise in ε' is observed. These results can be compared with the results obtained by other researchers (Sorichetti, 2011) for fatty acid methyl esters (FAME) before and after washing. They noted that as the temperature increases, more and more methanol reacts with the oil molecules; the new product obtained is a non-polar product with lower ε'. Thus, the concentration polar impurity in the form of methanol reduces with temperature and the dielectric value decreases. It should be noted that the frequency range (400–10,000 Hz) at which the ε' values were measured were much lower for the cited work than in present study.

3.4.4.2. Dielectric loss. For the oil–methanol-catalyst mixture, the values for ε'' decreased asymptotically with increasing frequency at all the temperatures (Fig. 6(d)). The ε'' is lower at 30 °C com-

pared to 45 and 60 °C (which were very close to each other). These results can be compared to literature data for FAME (Sorichetti, 2011). For an unwashed FAME product, the ε'' decreased with an increase in temperature. Unwashed FAME product contained unreacted methanol which increases the polarity and ultimately the ε''. However, with an increase in temperature, more and more methanol undergoes reaction and the amount of unreacted methanol decreases, decreasing the ε''. When the product was washed and was free of any polar impurities, the ε'' increased with temperature as expected.

4. Conclusions

Dielectric properties were both temperature and frequency dependent. Dielectric properties for oil (non-polar) were unaffected by the variations in frequency and temperature. Dielectric constant was higher for methanol than that for ethanol throughout the frequency range. Dielectric loss for methanol at higher frequencies (above 1000 MHz) was higher than ethanol and lower at lower frequencies. Ion solvation effect was observed for the alcohol–catalyst mixture. The dielectric properties were higher for methanol-catalyst mixture than that for ethanol-catalyst mixture throughout the frequency range. For the two heating methods, dielectric loss was much higher for the first scenario compared to the second scenario. The results of this study can be used to improve and optimize the microwave-assisted biodiesel production process as well as to develop numerical models for this process.

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